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Inventor(s)	Ozawa; Yoshihiko et al.

Power transmission apparatus

Abstract

During movement of a mass of a centrifugal clutch a radially inner position to a radially outer position and a resulting increase in torque transmitted from an input gear to an output shaft, a power transmission apparatus includes a first torque region where the apparatus restricts operation of a pressing assist cam and a second torque region where the apparatus allows operation of the pressing assist cam.

Inventors: Ozawa; Yoshihiko (Hamamatsu, JP), Chen; Han Hiong (Hamamatsu, JP), Yoshimoto; Katsu (Hamamatsu, JP)

Applicant: Kabushiki Kaisha F.C.C. (Hamamatsu, JP)

Family ID: 1000008820262

Assignee: KABUSHIKI KAISHA F.C.C. (Shizuoka, JP)

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Primary Examiner: Le; Huan

Attorney, Agent or Firm: Keating & Bennett, LLP

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a power transmission apparatus that is able to freely transmit a rotational force of an input to an output or cut off the rotational force.

2. Description of the Related Art

(2) Usually, a power transmission apparatus included in a motorcycle freely transmits a driving force of an engine to a transmission and a driving wheel or cuts off the driving force. Such a power transmission apparatus includes an input member connected to an engine, an output member connected to a transmission and a driving wheel, a clutch member connected to the output member, and a pressure member that is able to move close to or away from the clutch member. Bringing the pressure member close to the clutch member presses driving and driven clutch plates against each other so as to transmit power therebetween. Bringing the pressure member away from the clutch member releases a pressing force exerted on the driving and driven clutch plates so as to cut off transmission of the power therebetween.

(3) As disclosed, for example, in WO 2013/183588, a power transmission apparatus known in the art includes a centrifugal clutch means including a weight member that moves from a radially inner position of a groove to a radially outer position thereof with centrifugal force produced by rotation of a clutch housing and is thus able to press driving and driven clutch plates against each other. The power transmission apparatus known in the art is able to apply centrifugal force to the weight member in accordance with the rotation of the clutch housing caused by driving of an engine and is thus able to press the driving and driven clutch plates against each other so as to transmit a driving force of the engine to a wheel.

(4) The power transmission apparatus known in the art is provided with a pressing assist cam to increase a pressing force exerted on the driving and driven clutch plates when a rotational force received by an input member is transmittable to an output member. Accordingly, when a driver performs a clutch operation involving causing the driving and driven clutch plates to be pressed against each other, the power transmission apparatus known in the art is able to reduce an operating force so as to enable smooth power transmission.

SUMMARY OF THE INVENTION

(5) Unfortunately, during movement of the weight member of the centrifugal clutch means from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input member to the output member, the pressing assist cam of the power transmission apparatus known in the art may operate accidentally, for example, when the driver rides a clutch. This may result in sudden, unintentional power transmission when a vehicle starts to move, making it difficult for the vehicle to run smoothly.

(6) Preferred embodiments of the present invention provide power transmission apparatuses that are each able to prevent sudden, untimely power transmission caused by accidental operation of a pressing assist cam when a vehicle including a centrifugal clutch starts to move.

(7) A power transmission apparatus according to a preferred embodiment of the present invention

includes a clutch housing that rotates together with an input that rotates due to a driving force of an engine of a vehicle, the clutch housing including a plurality of driving clutch plates attached thereto, a clutch including a plurality of driven clutch plates attached thereto, the driven clutch plates being alternately arranged with the driving clutch plates attached to the clutch housing, the clutch being connected to an output that is able to rotate a wheel of the vehicle, a pressure applicator movable between an operating position where the driving and driven clutch plates are pressed against each other so as to enable transmission of the driving force of the engine to the wheel and a non-operating position where a pressing force exerted on the driving and driven clutch plates is released so as to cut off transmission of the driving force of the engine to the wheel, a centrifugal clutch including a mass movable from a radially inner position to a radially outer position with centrifugal force produced by rotation of the clutch housing, the centrifugal clutch being configured to, when the mass is located at the radially outer position, press the driving and driven clutch plates against each other so as to enable transmission of the driving force of the engine to the wheel and configured to, when the mass is located at the radially inner position, release the pressing force exerted on the driving and driven clutch plates so as to cut off transmission of the driving force of the engine to the wheel, and a pressing assist cam to increase the pressing force exerted on the driving and driven clutch plates when a rotational force received by the input is transmittable to the output. During movement of the mass of the centrifugal clutch from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input to the output, the power transmission apparatus includes a first torque region where the apparatus restricts operation of the pressing assist cam and a second torque region where the apparatus allows operation of the pressing assist cam.

(8) The clutch may include a first clutch connected to the output, and a second clutch having the driven clutch plates attached thereto. The pressing assist cam includes an inclined surface of the first clutch and an inclined surface of the pressure applicator that face each other.

(9) The power transmission apparatus may be configured to, in the first torque region, move the first clutch and the pressure applicator into abutment with each other so as to restrict operation of the pressing assist cam, and configured to, in the second torque region, move the first clutch and the pressure applicator away from each other so as to allow operation of the pressing assist cam.

(10) The centrifugal clutch may be configured to, in the first torque region, move the second clutch such that the first clutch and the pressure applicator are kept in abutment with each other, and configured to, in the second torque region, move the second clutch and the pressure applicator such that the first clutch and the pressure applicator are spaced away from each other.

(11) In the power transmission apparatus, a transition may be made from the first torque region to the second torque region in course of operation of the centrifugal clutch.

(12) According to a preferred embodiment of the present invention, during movement of the mass of the centrifugal clutch from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input to the output, the power transmission apparatus includes the first torque region where the apparatus restricts operation of the pressing assist cam, and the second torque region where the apparatus allows operation of the pressing assist cam. Consequently, this preferred embodiment is able to prevent sudden, untimely power transmission caused by accidental operation of the pressing assist cam when the vehicle including the centrifugal clutch starts to move.

(13) According to another preferred embodiment of the present invention, the clutch includes the first clutch connected to the output, and the second clutch having the driven clutch plates attached thereto. The pressing assist cam includes the inclined surface of the first clutch and the inclined surface of the pressure applicator that face each other. Consequently, with the first clutch and the pressure applicator, this preferred embodiment enables operation of the pressing assist cam.

(14) According to a further preferred embodiment of the present invention, the power transmission apparatus is configured to, in the first torque region, move the first clutch and the pressure

applicator into abutment with each other so as to restrict operation of the pressing assist cam, and is configured to, in the second torque region, move the first clutch and the pressure applicator away from each other so as to allow operation of the pressing assist cam. Consequently, this preferred embodiment is able to accurately and smoothly restrict operation of the pressing assist cam in the first torque region and allow operation of the pressing assist cam in the second torque region.

(15) According to an additional preferred embodiment of the present invention, the centrifugal clutch is configured to, in the first torque region, move the second clutch such that the first clutch and the pressure applicator are kept in abutment with each other, and is configured to, in the second torque region, move the second clutch and the pressure applicator such that the first clutch and the pressure applicator are spaced away from each other. Consequently, this preferred embodiment is able to restrict operation of the pressing assist cam in the first torque region and allow operation of the pressing assist cam in the second torque region by operating the centrifugal clutch.

(16) According to another preferred embodiment of the present invention, a transition is made from the first torque region to the second torque region in the course of operation of the centrifugal clutch. Consequently, this preferred embodiment is able to continuously and smoothly restrict operation of the pressing assist cam in the first torque region and allow operation of the pressing assist cam in the second torque region.

(17) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an external view of a power transmission apparatus according to a preferred embodiment of the present invention.

(2) FIG. 2 is a cross-sectional view taken along the line II-II in FIG. 1.

(3) FIG. 3 is a cross-sectional view taken along the line III-III in FIG. 1.

(4) FIG. 4 is a perspective view of a clutch housing of the power transmission apparatus.

(5) FIG. 5 is a three-view drawing of a first clutch member of the power transmission apparatus.

(6) FIG. 6 is a perspective view of the first clutch member.

(7) FIG. 7 is a three-view drawing of a second clutch member of the power transmission apparatus.

(8) FIG. 8 is a perspective view of the second clutch member.

(9) FIG. 9 is a three-view drawing of a pressure member of the power transmission apparatus.

(10) FIG. 10 is a perspective view of the pressure member.

(11) FIG. 11 is a longitudinal cross-sectional view of a centrifugal clutch of the power transmission apparatus.

(12) FIG. 12 is a partially cutaway perspective view of the centrifugal clutch.

(13) FIG. 13 is a three-view drawing of a holding member included in the centrifugal clutch.

(14) FIG. 14 is a three-view drawing of a supporting member included in the centrifugal clutch.

(15) FIG. 15 is a three-view drawing of a pressing member included in the centrifugal clutch.

(16) FIG. 16 is a four-view drawing of a weight member included in the centrifugal clutch.

(17) FIG. 17 is a cross-sectional view taken along the line XVII-XVII in FIG. 16.

(18) FIG. 18 is a plan view illustrating a state of the centrifugal clutch where each weight member is located at a radially inner position.

(19) FIG. 19 is a plan view illustrating a state of the centrifugal clutch where each weight member is located at a radially outer position.

(20) FIG. 20A is a schematic diagram describing how a pressing assist cam of the power transmission apparatus acts.

- (21) FIG. 20B is a schematic diagram describing how a back torque limiter cam of the power transmission apparatus acts.
- (22) FIG. 21 is a schematic diagram of a vehicle in which the power transmission apparatus is used.
- (23) FIG. 22 is a cross-sectional view illustrating a state of the power transmission apparatus where each weight member is located at the radially inner position.
- (24) FIG. 23 is a cross-sectional view illustrating a state of the power transmission apparatus where each weight member is located at an intermediate position between the radially inner position and the radially outer position.
- (25) FIG. 24 is a cross-sectional view illustrating a state of the power transmission apparatus where each weight member is located at the radially outer position.
- (26) FIG. 25 is a cross-sectional view illustrating a state of the power transmission apparatus where each weight member is located at the radially outer position and the pressure member is located at a non-operating position.
- (27) FIG. 26 is a graph illustrating relationships between engine rotation speeds and torques/assist cam positions for the power transmission apparatus according to the present preferred embodiment.
- (28) FIG. 27 is a graph illustrating relationships between engine rotation speeds and torques/assist cam positions for a power transmission apparatus known in the art.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(29) Preferred embodiments of the present invention will be described below in detail with reference to the drawings. As illustrated in FIG. 21, a power transmission apparatus K according to the present preferred embodiment is disposed in a vehicle so as to freely transmit a driving force of an engine E to a driving wheel T through a transmission M or cut off the driving force. As illustrated in FIGS. 1 to 17, the power transmission apparatus K includes a clutch housing 2 provided with an input gear 1 (which is an input) that rotates with the driving force of the engine E of the vehicle, an output shaft 3 (which is an output) connected to the transmission M, a clutch (which includes a first clutch member 4a and a second clutch member 4b); a pressure member 5 (which is a pressure applicator); a plurality of driving clutch plates 6; a plurality of driven clutch plates 7; a centrifugal clutch 9 including weight members 10 (which are masses); and an auxiliary clutch plate 17.

(30) The input gear 1 is rotatable around the output shaft 3 upon receiving a driving force (or a rotational force) transmitted from the engine E. The input gear 1 is connected to the clutch housing 2 with a fastener, such as a rivet. The clutch housing 2 is a cylindrical member with an opening defined at its right end in FIGS. 2 and 3 and is connected to the input gear 1. The clutch housing 2 is rotatable together with rotation of the input gear 1 with the driving force of the engine E.

(31) As illustrated in FIG. 4, the clutch housing 2 is circumferentially provided with a plurality of cut-outs 2a. The driving clutch plates 6 are fitted to the cut-outs 2A and thus attached to the clutch housing 2. The driving clutch plates 6 are each made of a substantially annular plate material. The driving clutch plates 6 are rotatable together with rotation of the clutch housing 2. The driving clutch plates 6 are slidable in an axial direction (which corresponds to a right-left direction in FIGS. 2 and 3).

(32) The driven clutch plates 7 are attached to the clutch (which includes the first clutch member 4a and the second clutch member 4b). The driven clutch plates 7 are arranged alternately with the driving clutch plates 6 attached to the clutch housing 2. The clutch is connected to the output shaft 3 (i.e., the output) that is able to rotate the driving wheel T through the transmission M of the vehicle. The clutch is provided by assembling two members, i.e., the first clutch member 4a and the second clutch member 4b, to each other.

(33) The output shaft 3 is inserted through an insertion hole (see FIGS. 5 and 6) defined in the center of the first clutch member 4a. A gear provided on the first clutch member 4a and a gear provided on the output shaft 3 mesh with each other so as to be connected to each other in the direction of rotation. As illustrated in FIGS. 5 and 6, the first clutch member 4a is provided with

inclined surfaces **4aa** each defining a pressing assist cam, and inclined surfaces **4ab** each defining a back torque limiter cam. The reference signs “**4ac**” in FIGS. **5** and **6** each indicate a boss provided with an insertion hole for a bolt B for connection between the first clutch member **4a** and a securing member **8**.

(34) As illustrated in FIGS. **7** and **8**, the second clutch member **4b** is an annular member provided with a flange **4bb**. The driven clutch plates **7** are spline-fitted to a spline-fitting portion **4ba** provided on the outer peripheral surface of the second clutch member **4b** and are thus attached to the second clutch member **4b**. As illustrated in FIGS. **2** and **3**, the pressure member **5** is assembled to the clutch (which includes the first clutch member **4a** and the second clutch member **4b**) such that the driving and driven clutch plates **6** and **7** are alternately stacked and secured between a flange **5c** of the pressure member **5** and the flange **4bb** of the second clutch member **4b**.

(35) As illustrated in FIGS. **9** and **10**, the pressure member **5** is a disk member provided with the flange **5c** extending along the peripheral edge of the pressure member **5**. The pressure member **5** is movable between an operating position where the driving and driven clutch plates **6** and **7** are pressed against each other so as to enable transmission of the driving force of the engine E to the wheel, and a non-operating position where a pressing force exerted on the driving and driven clutch plates **6** and **7** is released so as to cut off transmission of the driving force of the engine E to the wheel.

(36) More specifically, as illustrated in FIGS. **7** and **8**, the spline-fitting portion **4ba** provided on the second clutch member **4b** defines integral projections and depressions along substantially the entire outer peripheral side surface of the second clutch member **4b**. Fitting the driven clutch plates **7** to recessed grooves that define the spline-fitting portion **4ba** restricts movement of the driven clutch plates **7** in the direction of rotation while allowing movement of the driven clutch plates **7** in the axial direction relative to the second clutch member **4b**. The driven clutch plates **7** are rotatable together with the second clutch member **4b**.

(37) The driven clutch plates **7** are stacked alternately with the driving clutch plates **6**, making it possible to allow the clutch plates **6** and **7** adjacent to each other to be pressed against each other or release the pressing force exerted thereon. In other words, the clutch plates **6** and **7** are allowed to slide in the axial direction of the second clutch member **4b**. Engaging a clutch by pressing the clutch plates (**6a**, **6b**, **7a**, **7b**) against each other enables a rotational force of the clutch housing **2** to be transmitted to the output shaft **3** through the second clutch member **4b** and the first clutch member **4a**. Disengaging the clutch by releasing the pressing force exerted on the clutch plates (**6a**, **6b**, **7a**, **7b**) causes the first clutch member **4a** and the second clutch member **4b** to stop following the rotation of the clutch housing **2** such that no rotational force will be transmitted to the output shaft **3**.

(38) Accordingly, a state where the driving and driven clutch plates **6** and **7** are pressed against each other enables transmission of a rotational force (i.e., the driving force of the engine E), which is received by the clutch housing **2**, to the driving wheel (or the transmission M) through the output shaft **3** (i.e., the output), and a state where the driving and driven clutch plates **6** and **7** are not pressed against each other makes it possible to cut off transmission of the rotational force (or the driving force of the engine E), which is received by the clutch housing **2**, to the output shaft **3** (i.e., the output).

(39) As illustrated in FIGS. **9** and **10**, the pressure member **5** is circumferentially provided with a plurality of fitting holes **5d** (i.e., three fitting holes **5d** in the present preferred embodiment). A clutch spring S is fitted into each of the fitting holes **5d**. As illustrated in FIG. **2**, one end of each clutch spring S housed in the associated fitting hole **5d** is in abutment with the securing member **8**, such that each clutch spring S urges the pressure member **5** in a direction in which the driving and driven clutch plates **6** and **7** are to be pressed against each other. Operating a clutch actuator (not illustrated) makes it possible to press the driving and driven clutch plates **6** and **7** against each other or stop pressing the driving and driven clutch plates **6** and **7** against each other.

(40) In the present preferred embodiment, as illustrated in FIGS. 5, 6, 9, and 10, the first clutch member **4a** is provided with the inclined surfaces **4aa** and **4ab**, and the pressure member **5** is provided with inclined surfaces **5a** and **5b** respectively facing the inclined surfaces **4aa** and **4ab**. Specifically, each inclined surface **4aa** comes into abutment with the associated inclined surface **5a** so as to provide the pressing assist cam, and each inclined surface **4ab** comes into abutment with the associated inclined surface **5b** so as to provide the back torque limiter cam.

(41) Suppose that the rotation speed of the engine **E** is increased and the resulting rotational force received by the input gear **1** and the clutch housing **2** is transmittable to the output shaft **3** through the first clutch member **4a** and the second clutch member **4b** (which means that the weight members **10** are each located at a radially outer position). In this case, as illustrated in FIG. 20A, the rotational force is applied to the pressure member **5** in an a-direction, which exerts a force on the pressure member **5** in a c-direction in FIG. 20A under the action of each pressing assist cam. The pressure member **5** is thus moved in a direction in which its flange **5c** comes closer to the flange **4bb** of the second clutch member **4b** (i.e., leftward in FIGS. 2 and 3) so as to increase the pressing force exerted on the driving and driven clutch plates **6** and **7**.

(42) Suppose that the rotation of the output shaft **3** exceeds the rotation speed of the input gear **1** and the clutch housing **2**, resulting in a back torque. In this case, as illustrated in FIG. 20B, a rotational force is applied to the clutch member **4** in a b-direction, which moves the pressure member **5** in a d-direction in FIG. 20B under the action of each back torque limiter cam so as to release the pressing force exerted on the driving and driven clutch plates **6** and **7**. This makes it possible to prevent a malfunction in the power transmission apparatus **K** or a power source (i.e., the engine **E**) caused by the back torque.

(43) As illustrated in FIGS. 11 to 19, the centrifugal clutch **9** includes the weight members **10** each movable from a radially inner position (see FIG. 18) to the radially outer position (see FIG. 19) with centrifugal force produced by rotation of the clutch housing **2**. When the weight members **10** are each located at the radially outer position, the centrifugal clutch **9** presses the driving and driven clutch plates **6** and **7** against each other so as to enable transmission of the driving force of the engine **E** to the wheel (i.e., the driving wheel **T**). When the weight members **10** are each located at the radially inner position, the centrifugal clutch **9** releases the pressing force exerted on the driving and driven clutch plates **6** and **7** so as to cut off transmission of the driving force of the engine **E** to the wheel (i.e., the driving wheel **T**).

(44) Specifically, the centrifugal clutch **9** includes the weight members **10**, each of which is a substantially symmetrical polygonal member, a holding member **11** having a supporting member **13** attached thereto, a pressing member **12**, first spherical members **14**, second spherical members **15**, and urging members **16**, each of which is a coil spring. The holding member **11** and the pressing member **12** are each circumferentially provided with a plurality of protrusions. Similarly to the driving clutch plates **6**, the protrusions are fitted to the cut-outs **2a** of the clutch housing **2** such that the holding member **11** and the pressing member **12** are attached to the clutch housing **2**. Accordingly, the holding member **11** and the pressing member **12** are each movable in the axial direction of the clutch housing **2** and in engagement with the clutch housing **2** in the direction of rotation so as to be rotatable together with the clutch housing **2**.

(45) As illustrated in FIG. 16, each weight member **10** is a substantially symmetrical polygonal member including a first surface **X** and a second surface **Y**. As illustrated in FIGS. 16 and 17, each weight member **10** includes through holes **10a** passing through the weight member **10** from the first surface **X** to the second surface **Y**, insertion portions **10b** defined in the second surface **Y**, and a groove **10c** defined in the first surface **X**. As illustrated in FIGS. 18 and 19, the weight members **10** are each housed in an associated one of housing portions **11a** of the holding member **11**. When no centrifugal force is applied to the weight members **10**, the weight members **10** are each held at the radially inner position (see FIG. 18). Application of centrifugal force to the weight members **10** moves the weight members **10** outward against the urging force of the urging members **16** such that

each weight member **10** reaches the radially outer position (see FIG. **19**).

(46) The holding member **11** holds the weight members **10** such that each weight member **10** is movable between the radially inner position and the radially outer position. As illustrated in FIG. **13**, the holding member **11** is an annular member. The holding member **11** includes the housing portions **11a** which are arranged in the circumferential direction of the holding member **11** and in which the weight members **10** are to be housed, groove geometries **11b** defined in the housing portions **11a**, a pushing surface **11c**, a plurality of ribs **11d** located between the plurality of housing portions **11a** in the circumferential direction, and a plurality of pawl portions **11e** located radially outside the plurality of ribs **11d** and attached to the clutch housing **2**. Each housing portion **11a** has a recessed shape conforming to the shape and moving range of the associated weight member **10**. An inner peripheral wall surface **11aa** of each housing portion **11a** is allowed to abut against first ends of the associated urging members **16**. As shown in the side view of the holding member **11** in the middle of FIG. **13**, and in view of FIGS. **11** and **12**, a surface of the plurality of ribs **11d** facing the pressing member **12** (presser) and a surface of the plurality of pawl portions **11e** facing the pressing member **12** are flush with each other.

(47) The supporting member **13** is secured to a surface of the holding member **11**, which is provided with the housing portions **11a**. As illustrated in FIG. **14**, the supporting member **13** is provided with radially extending holding portions **13a**. The holding portions **13a** each conform to the groove **10c** of the associated weight member **10**, so that the weight members **10** are held by the holding member **11**. Specifically, the first surface X of each weight member **10** is centrally provided with the groove **10c** extending in a direction from the radially inner position to the radially outer position. Causing each holding portion **13a** to conform to the associated groove **10c** allows the holding member **11** to hold the weight members **10** such that each weight member **10** is movable in a radial direction (i.e., in a direction from the radially inner position to the radially outer position).

(48) Movement of each weight member **10** from the radially inner position to the radially outer position causes the pressing member **12** to move in a stacking direction of the driving and driven clutch plates **6** and **7** (i.e., rightward in FIGS. **2** and **3**) such that the driving and driven clutch plates **6** and **7** are pressed against each other. Specifically, as illustrated in FIG. **15**, the pressing member **12** is an annular member. The pressing member **12** includes inclined grooves **12a** (inclined surface) arranged in the circumferential direction of the pressing member **12**, groove geometries **12b** each defined at a position where the associated inclined groove **12a** is defined, a pushing surface **12c**, and a plurality of pawl portions **12e**.

(49) Each inclined groove **12a** is defined at a position corresponding to the position of the associated weight member **10**. Each inclined groove **12a** is inclined upward from its inner portion to its outer portion. Thus, when the clutch housing **2** is stationary, each weight member **10** is held at the radially inner position with the urging force of the associated urging members **16**. Rotation of the clutch housing **2** applies centrifugal force to the weight members **10** so as to move the weight members **10** along the inclined grooves **12a** inclined upward. This moves the pressing member **12** away from the holding member **11** (i.e., in the direction in which the driving and driven clutch plates **6** and **7** are to be pressed against each other).

(50) When the holding member **11** and the pressing member **12** are assembled to each other, with the weight members **10** interposed therebetween, each inclined groove **12a** is located at a position corresponding to the position of the associated weight member **10** as illustrated in FIGS. **11** and **12**. With centrifugal force, the weight members **10** each move along the inclined grooves **12a** from the radially inner position to the radially outer position, which moves the pressing member **12** in a direction indicated by the arrow in FIG. **11** (i.e., rightward in FIG. **11**). The pushing surface **12c** of the pressing member **12** thus pushes the driving and driven clutch plates **6** and **7** such that the driving and driven clutch plates **6** and **7** are pressed against each other. The resulting reaction force moves the holding member **11** in a direction opposite to the direction indicated by the arrow in

FIG. 11 (i.e., leftward in FIG. 11), so that the pushing surface **11c** of the holding member **11** presses the auxiliary clutch plate **17**.

(51) As illustrated in FIGS. **18** and **19**, the weight members **10** according to the present preferred embodiment are each housed in an associated one of the housing portions **11a** arranged in the circumferential direction of the holding member **11** such that the weight members **10** are movable radially. More than one urging member **16** (i.e., two urging members **16** in the present preferred embodiment) is disposed in the circumferential direction between the inner peripheral wall surface **11aa** (see FIG. 13) of each of the housing portions **11a** and an associated one of the weight members **10** so as to urge the associated weight member **10** from the radially outer position to the radially inner position. In this preferred embodiment, the inner peripheral wall surface **11aa** of each housing portion **11a** is a flat surface in abutment with the first ends of the associated urging members **16** such that the urging members **16** are stably securable.

(52) The weight members **10** according to the present preferred embodiment are each provided with the tunnel-shaped insertion portions **10b** which are openings defined in a surface of each weight member **10** facing the holding member **11** (i.e., the second surface Y in FIG. 17) and into which the urging members **16** are inserted such that the urging members **16** are attachable to the associated weight member **10**. The weight members **10**, with the urging members **16** inserted into the insertion portions **10b**, are housed in the housing portions **11a** of the holding member **11**. The urging members **16** are thus secured such that each urging member **16** is interposed between the inner peripheral wall surface **11aa** of the associated housing portion **11a** and the associated weight member **10**. The urging members **16** are disposed such that the first end of each urging member **16** is in abutment with the associated inner peripheral wall surface **11aa** and a second end of each urging member **16** is in abutment with an end wall surface **10ba** of the associated insertion portion **10b**, making it possible to urge each weight member **10** from the radially outer position to the radially inner position.

(53) Each first spherical member **14** is a steel ball attached to the associated weight member **10**. As illustrated in FIGS. **16** and **17**, each first spherical member **14** is partially protruded from a first opening **10aa** (i.e., a small-diameter opening in the first surface X) of the through hole **10a** defined in the associated weight member **10** and is in contact with a rolling contact surface of the pressing member **12** so as to be rollable thereon. Each second spherical member **15** is a steel ball attached to the associated weight member **10**. As illustrated in FIGS. **16** and **17**, each second spherical member **15** is partially protruded from a second opening **10ab** (i.e., a large-diameter opening in the second surface Y) of the through hole **10a** defined in the associated weight member **10** and is in contact with a rolling contact surface of the holding member **11** so as to be rollable thereon.

(54) As illustrated in FIG. 17, the through holes **10a** according to the present preferred embodiment are tapered such that the diameter of each through hole **10a** continuously increases from the first opening **10aa** (i.e., the small-diameter opening in the first surface X) to the second opening **10ab** (i.e., the large-diameter opening in the second surface Y). Disconnection of each first spherical member **14** from the associated through hole **10a** is prevented by the outer peripheral edge of one of the first opening **10aa** and the second opening **10ab** that has a smaller diameter (which is, in the present preferred embodiment, the first opening **10aa** in the first surface X). The first and second spherical members **14** and **15** according to the present preferred embodiment are spherical members having different diameters in accordance with the inner diameters of the through holes **10a**. The second spherical members **15** are larger in diameter than the first spherical members **14**. The small-diameter first spherical members **14** are each rollable while being in contact with the inner peripheral surface of the associated through hole **10a**, and at the same time, disconnection of the small-diameter first spherical members **14** is prevented by the small-diameter opening edges of the through holes **10a**.

(55) As illustrated in FIGS. **11** and **12**, disconnection of the second spherical members **15** is prevented by the rolling contact surface of the holding member **11**. Thus, disconnection of the

small-diameter first spherical members **14** is prevented by the small-diameter opening edges of the through holes **10a**, and disconnection of the large-diameter second spherical members **15** is prevented by the rolling contact surface of the holding member **11** while the second spherical members **15** are partially protruded from the large-diameter openings of the through holes **10a**. In the present preferred embodiment, the large-diameter second spherical members **15** are assembled to the weight members **10** such that the second spherical members **15** face the rolling contact surface of the holding member **11**. Alternatively, the second spherical members **15** may be assembled to the weight members **10** such that the second spherical members **15** face the rolling contact surface of the pressing member **12**. In this case, disconnection of the small-diameter first spherical members **14** is prevented by the small-diameter opening edges of the through holes **10a**, and disconnection of the large-diameter second spherical members **15** is prevented by the rolling contact surface of the pressing member **12** while the second spherical members **15** are partially protruded from the large-diameter openings of the through holes **10a**.

(56) As illustrated in FIG. **13**, the rolling contact surface of the holding member **11** (which is, in the present preferred embodiment, a surface of the holding member **11** on which the second spherical members **15** roll) includes the groove geometries **11b** each extending in the direction of movement of the associated weight member **10** (i.e., a direction connecting the radially inner position to the radially outer position). As illustrated in FIG. **15**, the rolling contact surface of the pressing member **12** (which is, in the present preferred embodiment, a surface of the pressing member **12** on which the first spherical members **14** roll) includes the groove geometries **12b** each extending in the direction of movement of the associated weight member **10** (i.e., a direction connecting the radially inner position to the radially outer position).

(57) As illustrated in FIGS. **16**, **18**, and **19**, the first and second spherical members **14** and **15** according to the present preferred embodiment are provided such that more than one first spherical member **14** and more than one second spherical member **15** are arranged in the circumferential direction of the holding member **11** (i.e., such that two first spherical members **14** and two second spherical members **15** are arranged in the width direction of each weight member **10** in the present preferred embodiment). In response to movement of the weight members **10**, the first spherical members **14** are movable along the groove geometries **12b** while rolling within the through holes **10a**, and the second spherical members **15** are movable along the groove geometries **11b** while rolling within the through holes **10a**.

(58) The auxiliary clutch plate **17** is an annular member different in diameter from the driving and driven clutch plates **6** and **7** (i.e., smaller in diameter than the driving and driven clutch plates **6** and **7** in the present preferred embodiment). As illustrated in FIGS. **2** and **3**, the output shaft **3** (i.e., the output) is inserted through a central opening **17a** of the auxiliary clutch plate **17** so as to be fitted thereto. The auxiliary clutch plate **17** includes a pushed surface **17b** facing the pushing surface **11c** of the holding member **11**.

(59) When the weight members **10** are each located at the radially outer position (i.e., when the driving and driven clutch plates **6** and **7** are pressed against each other), the auxiliary clutch plate **17** is able to transmit the driving force of the engine **E** to the output shaft **3** upon being pushed by the pushing surface **11c** of the holding member **11** and pressed against the pushing surface **11c**. When the weight members **10** are each located at the radially inner position (i.e., when the pressing force exerted on the driving and driven clutch plates **6** and **7** is released), the auxiliary clutch plate **17** is able to cut off transmission of the driving force of the engine **E** to the output shaft **3** upon being relieved of a pressing force applied thereto, owing to a decrease in pushing force exerted by the pushing surface **11c** of the holding member **11**.

(60) Upon movement of each weight member **10** to the radially outer position, each inclined groove **12a** functions as a cam so as to cause the holding member **11** and the pressing member **12** to move away from each other. Accordingly, the pushing surface **12c** of the pressing member **12** presses the driving and driven clutch plates **6** and **7** against each other, and the pushing surface **11c** of the

holding member **11** pushes the pushed surface **17b** of the auxiliary clutch plate **17** such that the pushed surface **17b** is pressed against the pushing surface **11c** so as to transmit the driving force of the engine E to the driving wheel T.

(61) In the power transmission apparatus K according to the present preferred embodiment, the through holes **10a** of the weight members **10** included in the centrifugal clutch **9** each have a tapered shape from the first opening **10aa** to the second opening **10ab**. Disconnection of each first spherical member **14** is prevented by the outer peripheral edge of one of the associated first opening **10aa** and the associated second opening **10ab** that has a smaller diameter. Accordingly, the present preferred embodiment is able to easily and accurately attach the first spherical members **14** to the weight members **10**, resulting in a reduction in manufacturing cost.

(62) The first and second spherical members **14** and **15** are spherical members having different diameters in accordance with the inner diameters of the through holes **10a**. The first and second spherical members **14** and **15** are rollable while being in contact with the inner peripheral surfaces of the through holes **10a**. Thus, during movement of the weight members **10**, the first and second spherical members **14** and **15** are rollable in a stable manner such that the weight members **10** move smoothly. Disconnection of the second spherical members **15** according to the present preferred embodiment is prevented by the rolling contact surface of the holding member **11** or the pressing member **12**. Consequently, the present preferred embodiment is able to facilitate preventing disconnection of the first and second spherical members **14** and **15**.

(63) The rolling contact surface of the holding member **11** or the pressing member **12** includes the groove geometries (**11b**, **12b**) extending in the direction of movement of the weight members **10**. Accordingly, the present preferred embodiment enables smoother movement of the weight members **10** while reliably preventing disconnection of the second spherical members **15** from the large-diameter openings and disconnection of the first spherical members **14** from the small-diameter openings.

(64) In addition, the weight members **10** according to the present preferred embodiment are each housed in an associated one of the housing portions **11a** arranged in the circumferential direction of the holding member **11** and are thus movable radially. More than one urging member **16** is disposed in the circumferential direction between the inner peripheral wall surface **11aa** of each of the housing portions **11a** and an associated one of the weight members **10** so as to urge the associated weight member **10** from the radially outer position to the radially inner position. Accordingly, the present preferred embodiment is able to accurately urge each weight member **10** from the radially outer position to the radially inner position, enabling stable movement of the weight members **10** in accordance with centrifugal force.

(65) The weight members **10** according to the present preferred embodiment are each provided with the insertion portions **10b** which are openings defined in the surface of each weight member **10** facing the holding member **11** and into which the urging members **16** are inserted such that the urging members **16** are attachable to the associated weight member **10**. This facilitates assembling the urging members **16** to the weight members **10**. The weight members **10** according to the present preferred embodiment are each provided with the groove **10c** extending in a direction from the radially inner position to the radially outer position. The holding member **11** (or specifically, the supporting member **13** secured to the holding member **11** so as to be integral therewith) is provided with the holding portions **13a** each conforming to the associated groove **10c** and holding the associated weight member **10**. Accordingly, the present preferred embodiment enables stable movement of the weight members **10**.

(66) The centrifugal clutch **9** according to the present preferred embodiment includes the first spherical members **14** that are partially protruded from the first openings **10aa** of the through holes **10a** defined in the weight members **10** and are in contact with the rolling contact surface (or the groove geometries **12b**) of the pressing member **12** so as to be rollable thereon, and the second spherical members **15** that are partially protruded from the second openings **10ab** of the through

holes **10a** defined in the weight members **10** and are in contact with the rolling contact surface (or the groove geometries **11b**) of the holding member **11** so as to be rollable thereon. Accordingly, the present preferred embodiment enables more stable movement of the weight members **10**.

(67) The holding member **11** or the pressing member **12**, in particular, includes the groove geometries (**11b**, **12b**) extending in the direction of movement of the weight members **10**. The groove geometries (**11b**, **12b**) define the rolling contact surface for the first spherical members **14** or the second spherical members **15** so as to enable smoother movement of the weight members **10**. The first and second spherical members **14** and **15** according to the present preferred embodiment are provided such that more than one first spherical member **14** and more than one second spherical member **15** are arranged in the circumferential direction of the holding member **11** (i.e., the width direction of each weight member **10**). Accordingly, the present preferred embodiment enables more stable movement of the weight members **10**.

(68) In the course of movement of each weight member **10** of the centrifugal clutch **9** from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input gear **1** (i.e., the input) to the output shaft **3** (i.e., the output), the power transmission apparatus **K** according to the present preferred embodiment includes, as illustrated in FIG. **26**, a first torque region $\alpha 1$ where the apparatus restricts operation of the pressing assist cams (which are cams including the inclined surfaces **4aa** and **5a**) and a second torque region $\alpha 2$ where the apparatus allows operation of the pressing assist cams.

(69) Specifically, as illustrated in FIGS. **5** and **6**, a portion of a surface of the first clutch member **4a** according to the present preferred embodiment, which faces the pressure member **5**, defines an abutment surface **4ad**. As illustrated in FIGS. **9** and **10**, a portion of a surface of the pressure member **5**, which faces the first clutch member **4a**, defines an abutment surface **5e**. With the first clutch member **4a**, the second clutch member **4b**, and the pressure member **5** assembled to each other (and with no torque being transmitted from the input gear **1** (i.e., the input) to the output shaft **3** (i.e., the output)), the abutment surface **4ad** and the abutment surface **5e** are in abutment with each other as illustrated in FIGS. **2** and **3**.

(70) With the abutment surface **4ad** and the abutment surface **5e** in abutment with each other as mentioned above (i.e., when the apparatus is in the first torque region $\alpha 1$ corresponding to a time period between a time $t1$ and a time $t2$ in FIG. **26**), in the course of movement of each weight member **10** of the centrifugal clutch **9** from the radially inner position (see FIG. **22**) to an intermediate position (see FIG. **23**) and a resulting increase in torque transmitted from the input gear **1** (i.e., the input) to the output shaft **3** (i.e., the output), the first clutch member **4a** and the pressure member **5** are not allowed to move relative to each other, which restricts operation of the pressing assist cams.

(71) Each weight member **10** of the centrifugal clutch **9** then further moves from the intermediate position (see FIG. **23**) to the radially outer position (see FIG. **24**), and the driving and driven clutch plates **6** and **7** are pushed by the flange **4bb** of the second clutch member **4b** and thus pressed against each other. When the pushing force of the flange **4bb** is equal to or greater than the urging force of the clutch springs **S**, the second clutch member **4b** and the pressure member **5** are moved in the axial direction (i.e., rightward in FIGS. **2** and **3**) relative to the first clutch member **4a**, so that the abutment surface **4ad** of the first clutch member **4a** and the abutment surface **5e** of the pressure member **5** are spaced away from each other. FIG. **25** illustrates a state in which the weight members **10** are each located at the radially outer position and the pressure member **5** is located at the non-operating position (i.e., a clutch-disengaged state).

(72) With the abutment surface **4ad** and the abutment surface **5e** spaced away from each other as mentioned above (i.e., when the apparatus is in the second torque region $\alpha 2$ corresponding to a time period subsequent to the time $t2$ in FIG. **26**), in the course of movement of each weight member **10** of the centrifugal clutch **9** from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input gear **1** (i.e., the input) to the output shaft **3**

(i.e., the output), the first clutch member **4a** and the pressure member **5** are allowed to move relative to each other, which allows operation of the pressing assist cams.

(73) In other words, the present preferred embodiment involves, in the first torque region $\alpha 1$, moving the abutment surface **4ad** of the first clutch member **4a** and the abutment surface **5e** of the pressure member **5** into abutment with each other so as to restrict operation of the pressing assist cams, and involves, in the second torque region $\alpha 2$, moving the abutment surface **4ad** of the first clutch member **4a** and the abutment surface **5e** of the pressure member **5** away from each other so as to allow operation of the pressing assist cams.

(74) In the present preferred embodiment, the centrifugal clutch **9** is configured to, in the first torque region $\alpha 1$, move the second clutch member **4b** without moving the first clutch member **4a** such that the first clutch member **4a** and the pressure member **5** are kept in abutment with each other, and configured to, in the second torque region $\alpha 2$, move the second clutch member **4b** and the pressure member **5** such that the abutment surface **4ad** of the first clutch member **4a** and the abutment surface **5e** of the pressure member **5** are spaced away from each other. The present preferred embodiment, in particular, involves making a transition from the first torque region $\alpha 1$ to the second torque region $\alpha 2$ in the course of operation of the centrifugal clutch **9**.

(75) For example, suppose that a power transmission apparatus known in the art includes a first clutch member **4a** and a pressure member **5** that do not come into abutment with each other and includes no first torque region $\alpha 1$ (but includes only a torque region B). In this case, as illustrated in FIG. 27, pressing assist cams unfortunately operate upon generation of a torque transmitted from an input gear **1** (i.e., an input) to an output shaft **3** (i.e., an output), which is caused by movement of each weight member **10** of a centrifugal clutch **9** from a radially inner position to a radially outer position. This results in sudden, unintentional power transmission when a vehicle starts to move, making it difficult for the vehicle to run smoothly.

(76) In the course of movement of each weight member **10** of the centrifugal clutch **9** from the radially inner position to the radially outer position and a resulting increase in torque transmitted from the input gear **1** (i.e., the input) to the output shaft **3** (i.e., the output), the apparatus according to the present preferred embodiment includes the first torque region $\alpha 1$ where the apparatus restricts operation of the pressing assist cams, and the second torque region $\alpha 2$ where the apparatus allows operation of the pressing assist cams. Accordingly, the present preferred embodiment is able to prevent sudden, untimely power transmission caused by accidental operation of the pressing assist cams when the vehicle including the centrifugal clutch **9** starts to move.

(77) The clutch according to the present preferred embodiment includes the first clutch member **4a** connected to the output shaft **3** (i.e., the output), and the second clutch member **4b** having the driven clutch plates **7** attached thereto. The pressing assist cams each include an associated one of the inclined surfaces **4aa** of the first clutch member **4a** and an associated one of the inclined surfaces **5a** of the pressure member **5** that face each other. Accordingly, with the first clutch member **4a** and the pressure member **5**, the present preferred embodiment enables operation of the pressing assist cams.

(78) In the first torque region $\alpha 1$, the first clutch member **4a** and the pressure member **5** are in abutment with each other so as to restrict operation of the pressing assist cams. In the second torque region $\alpha 2$, the first clutch member **4a** and the pressure member **5** are spaced away from each other so as to allow operation of the pressing assist cams. Accordingly, the present preferred embodiment is able to accurately and smoothly restrict operation of the pressing assist cams in the first torque region $\alpha 1$ and allow operation of the pressing assist cams in the second torque region $\alpha 2$.

(79) In the first torque region $\alpha 1$, the centrifugal clutch **9** moves the second clutch member **4b** without moving the first clutch member **4a** such that the first clutch member **4a** and the pressure member **5** are kept in abutment with each other. In the second torque region $\alpha 2$, the centrifugal clutch **9** moves the second clutch member **4a** and the pressure member **5** such that the first clutch member **4a** and the pressure member **5** are spaced away from each other. Accordingly, the present

preferred embodiment is able to restrict operation of the pressing assist cams in the first torque region $\alpha 1$ and allow operation of the pressing assist cams in the second torque region $\alpha 2$ by operating the centrifugal clutch **9**.

(80) The present preferred embodiment, in particular, involves making a transition from the first torque region $\alpha 1$ to the second torque region $\alpha 2$ in the course of operation of the centrifugal clutch **9** (i.e., in the course of movement of each weight member **10** from the radially inner position to the radially outer position). Accordingly, the present preferred embodiment is able to continuously and smoothly restrict operation of the pressing assist cams in the first torque region $\alpha 1$ and allow operation of the pressing assist cams in the second torque region $\alpha 2$.

(81) Although the present preferred embodiment has been described thus far, the present invention is not limited to this preferred embodiment. In one example, the apparatus may include no back torque limiter cam (which includes the inclined surfaces **4ab** and **5b**). In another example, the apparatus may include no auxiliary clutch plate **17**. In still another example, the centrifugal clutch **9** may be in any other form (such as one including a weight member made of a steel ball). The power transmission apparatuses according to preferred embodiments of the present invention may find applications as various multiple-plate clutch type power transmission apparatuses for, for example, motorcycles, automobiles, three-wheel or four-wheel buggies, or general purpose machines.

(82) During movement of each weight member of a centrifugal clutch from a radially inner position to a radially outer position and a resulting increase in torque transmitted from an input to an output, a power transmission apparatus includes a first torque region where the apparatus restricts operation of pressing assist cams and a second torque region where the apparatus allows operation of the pressing assist cams. Such a power transmission apparatus may find applications involving, for example, change(s) in external shape or addition of other function(s).

(83) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A power transmission apparatus comprising: a clutch housed in a clutch housing that rotates together with an input that rotates due to a driving force of an engine of a vehicle, the clutch housing including a plurality of driving clutch plates attached thereto, the clutch being connected to an output that is able to rotate a wheel of the vehicle; a pressure applicator having at least one driven clutch plate attached thereto, the driven clutch plate being alternately arranged with the driving clutch plates, the pressure applicator movable between an operating position where the driving and driven clutch plates are pressed against each other so as to enable transmission of the driving force of the engine to the wheel and a non-operating position where a pressing force exerted on the driving and driven clutch plates is released so as to cut off transmission of the driving force of the engine to the wheel; and a centrifugal clutch including a mass movable from a radially inner position to a radially outer position due to centrifugal force produced by rotation of the clutch housing, the centrifugal clutch being configured to, when the mass is located at the radially outer position, press the driving and driven clutch plates against each other so as to enable transmission of the driving force of the engine to the wheel and configured to, when the mass is located at the radially inner position, release the pressing force exerted on the driving and driven clutch plates so as to cut off transmission of the driving force of the engine to the wheel; wherein the centrifugal clutch includes: a holder including a plurality of housing portions in a circumferential direction housing the mass and a plurality of holder pawl portions on an outer peripheral surface thereof and attached to the clutch housing, and holding the mass such that the mass is movable between the radially inner position and the radially outer position; and a presser to

press the driving and driven clutch plates against each other by moving in a stacking direction of the driving and driven clutch plates in response to movement of the mass from the radially inner position to the radially outer position; the presser is located on an opposite side of the holder across the mass and between the mass and the driving clutch plates and the driven clutch plate in an axial direction of the output member; and some of the holder pawl portions at least partially overlap with the housing portion when the holder pawl portions are respectively seen in a radial direction of the output, and others of the holder pawl portions at least partially do not overlap with the housing portion when the holder pawl portions are respectively seen in the radial direction of the output.

2. The power transmission apparatus according to claim 1, wherein the some of the holder pawl portions overlap with the housing portion in one half or more of a length of the holder pawl portion in the circumferential direction when the holder pawl portions are respectively seen in the radial direction of the output, and the others of the holder pawl portions do not overlap with the housing portion in one half or more of a length of the holder pawl portion in the circumferential direction when the holder pawl portions are respectively seen in the radial direction of the output.

3. The power transmission apparatus according to claim 1, wherein the presser includes a plurality of presser pawl portions on an outer peripheral surface thereof and attached to cut-outs of the clutch housing; the holder pawl portions are attached to the cut-outs of the clutch housing; and mutually opposing surfaces of the holder pawl portions and the presser pawl portions are parallel to each other.

4. The power transmission apparatus according to claim 1, wherein one surface of the holder pawl portions and another surface of the holder pawl portions in the axial direction of the output are parallel to each other.

5. The power transmission apparatus according to claim 4, wherein the presser includes a plurality of presser pawl portions on an outer peripheral surface thereof and attached to cut-outs of the clutch housing; and one surface of the holder pawl portions and another surface of the holder pawl portions in the axial direction of the output and one surface of the presser pawl portions and another surface of the presser pawl portions in the axial direction of the output are parallel to each other.

6. The power transmission apparatus according to claim 4, wherein the housing portion includes a contact surface that comes into contact with a surface of the mass on one side in the axial direction of the output; and the contact surface and the one surface of the holder pawl portions and the other surface of the holder pawl portions in the axial direction of the output are parallel to each other.

7. The power transmission apparatus according to claim 4, wherein the centrifugal clutch includes a spring to urge the mass to a radially inner position; and an axis of the spring and the one surface of the holder pawl portions and the other surface of the holder pawl portions in the axial direction of the output are parallel to each other.
